# Supplementary Material

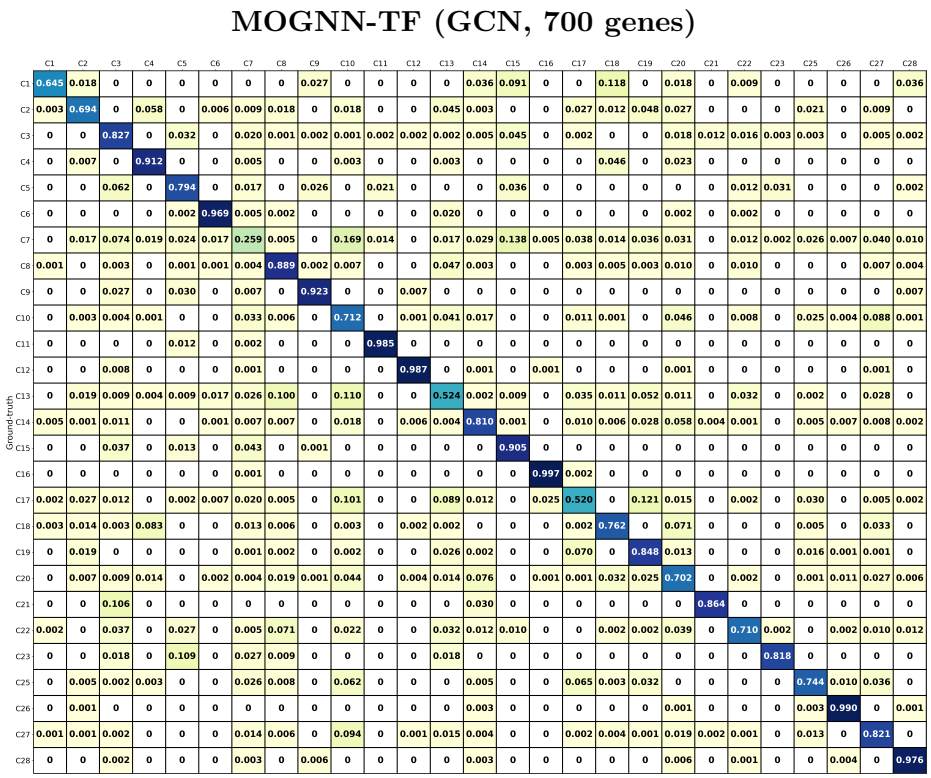
MOGNN-TF: a heterogeneous graph neural network integrating transcription-factor regulation for pan-cancer molecular subtype classification

## Subtype legend

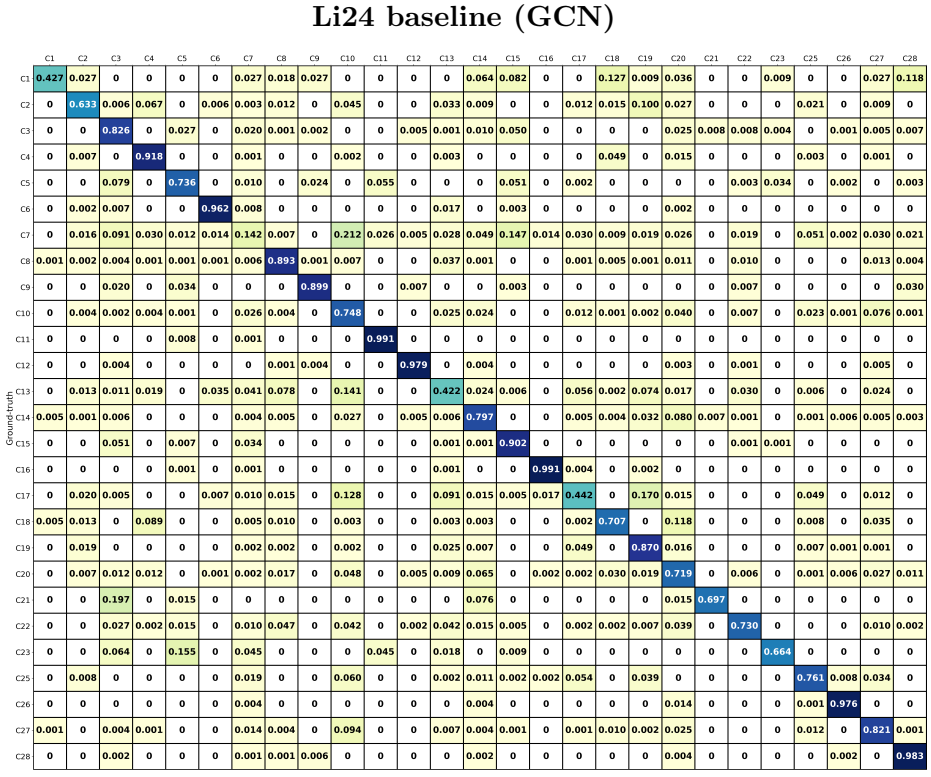
The iCluster pan-cancer subtypes (labels C1–C28; C24, LAML, is absent from the per-class evaluation table, so numbering skips from C23 to C25) are summarised below. Numbering follows Hoadley et al.

**Subtype legend.** iCluster subtype legend.

| Label | Dominant tumour type / characterisation |
|---|---|
| C1 | BRCA (luminal-related) |
| C2 | BRCA, HER2/ERBB2 amplified (also BLCA, STAD) |
| C3 | Mesenchymal / immune-infiltrated |
| C4 | Pan-GI (CRC; COAD/READ, chromosomal instability) |
| C5 | CNS / endocrine |
| C6 | OV (ovarian) |
| C7 | Mixed, chr9 deletion |
| C8 | UCEC (endometrial) |
| C9 | ACC/KICH, hypodiploid |
| C10 | Pan-SCC, 9p deletion (LUSC frequent) |
| C11 | LGG, IDH1-mutant |
| C12 | THCA (thyroid) |
| C13 | Mixed, chr8 deletion (highly aneuploid) |
| C14 | LUAD (lung adenocarcinoma) |
| C15 | SKCM/UVM (melanoma) |
| C16 | PRAD (prostate) |
| C17 | BRCA, chr8q amplified |
| C18 | Pan-GI, MSI (STAD/COAD) |
| C19 | BRCA (luminal) |
| C20 | Mixed, stromal/immune |
| C21 | DLBC (lymphoma) |
| C22 | TGCT (testicular germ-cell) |
| C23 | GBM/LGG, IDH1 wild-type |
| C25 | Pan-SCC, chr11 amplified |
| C26 | LIHC (hepatocellular) |
| C27 | Pan-SCC, HPV-associated (CESC, HPV+ HNSC) |
| C28 | Pan-kidney (KIRC, KIRP) |




(a) MOGNN-TF (GCN backbone, 700 selected genes).




(b) Li24 baseline (Li et al., GCN backbone).

Figure S1: Row-normalised confusion matrices on the 27-class iCluster pan-cancer taxonomy, averaged over the eleven seeds on the test set, for the proposed MOGNN-TF (GCN, 700 genes) (top) and the Li24 baseline (Li et al., GCN) (bottom). Each row is normalised by its class support, so the diagonal reports per-class recall. The residual off-diagonal mass concentrates on a small number of low-support, biologically adjacent clusters (e.g. the BRCA-related C1/C17/C19 group and the chr8/chr9-deletion clusters C7/C13).

Table S1: Per-class precision (P), recall (R) and F1 for the five compared models, mean ± s.d. over eleven seeds on the test set. Best value per metric within each class in bold.

| Class | Li24 (GAT) | | | Li24 (GCN) | | | MoGCN | | | MOGNN-TF (GAT) | | | MOGNN-TF (GCN) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **P** | **R** | **F1** | **P** | **R** | **F1** | **P** | **R** | **F1** | **P** | **R** | **F1** | **P** | **R** | **F1** |
| C1 | **0.863** ± 0.162 | 0.445 ± 0.192 | 0.562 ± 0.191 | 0.835 ± 0.175 | 0.427 ± 0.215 | 0.533 ± 0.217 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.794 ± 0.141 | 0.618 ± 0.183 | 0.670 ± 0.149 | 0.833 ± 0.131 | **0.645** ± 0.192 | **0.709** ± 0.142 |
| C2 | 0.727 ± 0.161 | 0.530 ± 0.106 | 0.609 ± 0.117 | 0.722 ± 0.124 | 0.633 ± 0.106 | 0.668 ± 0.091 | 0.343 ± 0.167 | 0.079 ± 0.052 | 0.122 ± 0.072 | 0.707 ± 0.088 | **0.794** ± 0.076 | **0.742** ± 0.051 | **0.765** ± 0.088 | 0.718 ± 0.103 | 0.737 ± 0.077 |
| C3 | 0.792 ± 0.051 | 0.828 ± 0.047 | 0.808 ± 0.033 | 0.795 ± 0.034 | 0.826 ± 0.063 | 0.809 ± 0.039 | 0.755 ± 0.040 | **0.873** ± 0.049 | 0.809 ± 0.037 | 0.815 ± 0.057 | 0.827 ± 0.052 | **0.819** ± 0.038 | **0.820** ± 0.045 | 0.819 ± 0.055 | 0.818 ± 0.039 |
| C4 | 0.874 ± 0.046 | 0.904 ± 0.029 | 0.888 ± 0.028 | 0.875 ± 0.026 | 0.918 ± 0.035 | 0.896 ± 0.023 | 0.864 ± 0.028 | **0.938** ± 0.022 | 0.899 ± 0.021 | **0.913** ± 0.033 | 0.899 ± 0.050 | 0.905 ± 0.030 | 0.891 ± 0.056 | 0.924 ± 0.033 | **0.906** ± 0.027 |
| C5 | 0.824 ± 0.057 | 0.767 ± 0.057 | 0.792 ± 0.036 | 0.845 ± 0.084 | 0.736 ± 0.088 | 0.784 ± 0.073 | **0.885** ± 0.044 | 0.672 ± 0.072 | 0.762 ± 0.048 | 0.853 ± 0.089 | 0.758 ± 0.057 | 0.799 ± 0.040 | 0.833 ± 0.059 | **0.782** ± 0.071 | **0.803** ± 0.040 |
| C6 | 0.929 ± 0.026 | 0.965 ± 0.028 | 0.947 ± 0.018 | **0.952** ± 0.029 | 0.962 ± 0.021 | 0.957 ± 0.019 | 0.944 ± 0.031 | **0.982** ± 0.014 | **0.962** ± 0.021 | 0.940 ± 0.027 | 0.957 ± 0.027 | 0.948 ± 0.022 | 0.947 ± 0.030 | 0.959 ± 0.035 | 0.952 ± 0.026 |
| C7 | 0.300 ± 0.110 | 0.196 ± 0.095 | 0.229 ± 0.085 | 0.265 ± 0.104 | 0.142 ± 0.060 | 0.180 ± 0.069 | 0.312 ± 0.150 | 0.093 ± 0.037 | 0.143 ± 0.059 | 0.269 ± 0.083 | 0.226 ± 0.073 | 0.244 ± 0.075 | **0.321** ± 0.107 | **0.235** ± 0.075 | **0.262** ± 0.068 |
| C8 | 0.865 ± 0.040 | 0.886 ± 0.036 | 0.874 ± 0.022 | 0.880 ± 0.041 | 0.893 ± 0.037 | 0.886 ± 0.026 | 0.836 ± 0.052 | **0.899** ± 0.029 | 0.865 ± 0.030 | 0.901 ± 0.027 | 0.885 ± 0.047 | **0.892** ± 0.027 | **0.903** ± 0.040 | 0.878 ± 0.029 | 0.890 ± 0.025 |
| C9 | **0.910** ± 0.034 | 0.906 ± 0.071 | 0.906 ± 0.038 | 0.891 ± 0.050 | 0.899 ± 0.044 | 0.895 ± 0.040 | 0.895 ± 0.041 | **0.956** ± 0.040 | **0.924** ± 0.027 | 0.888 ± 0.068 | 0.916 ± 0.055 | 0.900 ± 0.041 | 0.905 ± 0.041 | 0.906 ± 0.048 | 0.904 ± 0.027 |
| C10 | 0.654 ± 0.072 | 0.701 ± 0.044 | 0.674 ± 0.044 | 0.669 ± 0.041 | 0.748 ± 0.055 | **0.705** ± 0.032 | 0.555 ± 0.024 | **0.774** ± 0.028 | 0.646 ± 0.022 | 0.662 ± 0.072 | 0.703 ± 0.079 | 0.675 ± 0.027 | **0.681** ± 0.037 | 0.707 ± 0.069 | 0.691 ± 0.029 |
| C11 | **0.973** ± 0.022 | 0.985 ± 0.009 | **0.979** ± 0.010 | 0.949 ± 0.033 | 0.991 ± 0.008 | 0.970 ± 0.018 | 0.869 ± 0.036 | **0.998** ± 0.005 | 0.929 ± 0.021 | 0.959 ± 0.023 | 0.983 ± 0.026 | 0.971 ± 0.017 | 0.965 ± 0.025 | 0.983 ± 0.024 | 0.973 ± 0.015 |
| C12 | **0.982** ± 0.014 | 0.978 ± 0.020 | 0.980 ± 0.009 | 0.979 ± 0.016 | 0.979 ± 0.014 | 0.979 ± 0.009 | 0.962 ± 0.024 | 0.980 ± 0.014 | 0.971 ± 0.014 | 0.979 ± 0.015 | **0.985** ± 0.019 | 0.982 ± 0.010 | 0.981 ± 0.021 | **0.985** ± 0.013 | **0.982** ± 0.008 |
| C13 | 0.440 ± 0.034 | 0.379 ± 0.114 | 0.398 ± 0.071 | **0.460** ± 0.058 | 0.422 ± 0.078 | 0.438 ± 0.063 | 0.435 ± 0.084 | 0.318 ± 0.067 | 0.363 ± 0.064 | 0.456 ± 0.090 | **0.506** ± 0.156 | **0.469** ± 0.091 | 0.442 ± 0.083 | 0.487 ± 0.100 | 0.458 ± 0.070 |
| C14 | 0.778 ± 0.036 | 0.789 ± 0.091 | 0.780 ± 0.045 | 0.768 ± 0.048 | 0.797 ± 0.066 | 0.780 ± 0.033 | **0.801** ± 0.019 | **0.810** ± 0.020 | **0.805** ± 0.010 | 0.779 ± 0.056 | 0.801 ± 0.044 | 0.787 ± 0.024 | 0.795 ± 0.038 | 0.804 ± 0.047 | 0.798 ± 0.024 |
| C15 | 0.797 ± 0.029 | 0.890 ± 0.058 | 0.840 ± 0.033 | 0.787 ± 0.043 | 0.902 ± 0.038 | 0.840 ± 0.033 | 0.775 ± 0.036 | **0.949** ± 0.032 | **0.853** ± 0.027 | 0.800 ± 0.046 | 0.881 ± 0.055 | 0.836 ± 0.027 | **0.807** ± 0.040 | 0.906 ± 0.041 | 0.852 ± 0.015 |
| C16 | 0.981 ± 0.013 | 0.995 ± 0.009 | 0.988 ± 0.007 | 0.984 ± 0.012 | 0.991 ± 0.009 | 0.988 ± 0.007 | 0.982 ± 0.013 | 0.995 ± 0.005 | 0.989 ± 0.008 | 0.983 ± 0.018 | 0.995 ± 0.007 | 0.989 ± 0.009 | **0.986** ± 0.011 | **0.997** ± 0.005 | **0.991** ± 0.006 |
| C17 | 0.466 ± 0.106 | 0.337 ± 0.091 | 0.384 ± 0.080 | 0.525 ± 0.087 | 0.442 ± 0.049 | 0.476 ± 0.049 | 0.467 ± 0.080 | 0.140 ± 0.032 | 0.212 ± 0.035 | 0.447 ± 0.115 | 0.410 ± 0.118 | 0.418 ± 0.099 | **0.531** ± 0.101 | **0.467** ± 0.086 | **0.491** ± 0.080 |
| C18 | 0.707 ± 0.065 | 0.689 ± 0.078 | 0.693 ± 0.036 | 0.757 ± 0.058 | 0.707 ± 0.045 | 0.730 ± 0.046 | 0.671 ± 0.055 | 0.648 ± 0.067 | 0.657 ± 0.044 | 0.756 ± 0.092 | **0.805** ± 0.086 | **0.773** ± 0.037 | **0.780** ± 0.058 | 0.770 ± 0.115 | 0.769 ± 0.060 |
| C19 | 0.793 ± 0.038 | 0.843 ± 0.054 | 0.816 ± 0.035 | 0.815 ± 0.025 | 0.870 ± 0.048 | **0.840** ± 0.024 | 0.683 ± 0.015 | **0.927** ± 0.033 | 0.786 ± 0.014 | 0.832 ± 0.041 | 0.839 ± 0.044 | 0.834 ± 0.023 | **0.841** ± 0.026 | 0.843 ± 0.063 | **0.841** ± 0.032 |
| C20 | 0.698 ± 0.059 | **0.722** ± 0.055 | 0.706 ± 0.024 | 0.761 ± 0.045 | 0.719 ± 0.051 | 0.737 ± 0.029 | 0.710 ± 0.023 | 0.701 ± 0.035 | 0.705 ± 0.026 | **0.796** ± 0.046 | 0.690 ± 0.049 | 0.737 ± 0.022 | 0.776 ± 0.048 | 0.713 ± 0.067 | **0.740** ± 0.031 |
| C21 | 0.809 ± 0.167 | 0.606 ± 0.187 | 0.665 ± 0.162 | **0.817** ± 0.185 | 0.697 ± 0.180 | 0.723 ± 0.085 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.785 ± 0.196 | 0.818 ± 0.174 | 0.788 ± 0.161 | 0.766 ± 0.142 | **0.879** ± 0.151 | **0.799** ± 0.084 |
| C22 | 0.781 ± 0.088 | **0.764** ± 0.074 | 0.768 ± 0.054 | 0.822 ± 0.091 | 0.730 ± 0.086 | 0.770 ± 0.070 | **0.905** ± 0.080 | 0.703 ± 0.090 | **0.786** ± 0.055 | 0.811 ± 0.089 | 0.759 ± 0.086 | 0.778 ± 0.052 | 0.804 ± 0.071 | 0.757 ± 0.074 | 0.775 ± 0.036 |
| C23 | 0.753 ± 0.132 | 0.682 ± 0.087 | 0.706 ± 0.062 | 0.739 ± 0.095 | 0.664 ± 0.175 | 0.690 ± 0.128 | **0.887** ± 0.097 | 0.455 ± 0.151 | 0.587 ± 0.120 | 0.682 ± 0.091 | **0.755** ± 0.163 | **0.708** ± 0.109 | 0.715 ± 0.084 | 0.700 ± 0.134 | 0.696 ± 0.061 |
| C25 | 0.769 ± 0.115 | 0.636 ± 0.134 | 0.693 ± 0.117 | 0.803 ± 0.067 | **0.761** ± 0.066 | 0.779 ± 0.046 | 0.273 ± 0.410 | 0.006 ± 0.009 | 0.013 ± 0.018 | 0.784 ± 0.067 | 0.693 ± 0.072 | 0.733 ± 0.049 | **0.824** ± 0.062 | 0.748 ± 0.065 | **0.783** ± 0.054 |
| C26 | **0.960** ± 0.022 | 0.983 ± 0.013 | **0.971** ± 0.012 | **0.960** ± 0.026 | 0.976 ± 0.022 | 0.967 ± 0.012 | 0.922 ± 0.033 | 0.996 ± 0.010 | 0.957 ± 0.020 | 0.942 ± 0.040 | 0.980 ± 0.020 | 0.960 ± 0.018 | 0.947 ± 0.041 | **0.997** ± 0.006 | **0.971** ± 0.021 |
| C27 | 0.763 ± 0.035 | 0.789 ± 0.051 | 0.775 ± 0.029 | 0.799 ± 0.030 | 0.821 ± 0.048 | **0.809** ± 0.023 | 0.710 ± 0.026 | **0.845** ± 0.025 | 0.771 ± 0.023 | **0.820** ± 0.057 | 0.774 ± 0.070 | 0.793 ± 0.028 | 0.800 ± 0.054 | 0.824 ± 0.053 | **0.809** ± 0.027 |
| C28 | 0.965 ± 0.015 | 0.982 ± 0.014 | 0.973 ± 0.006 | 0.959 ± 0.016 | **0.983** ± 0.013 | 0.970 ± 0.007 | 0.943 ± 0.019 | 0.977 ± 0.014 | 0.959 ± 0.011 | **0.969** ± 0.018 | 0.980 ± 0.010 | **0.975** ± 0.008 | 0.967 ± 0.022 | **0.983** ± 0.012 | **0.975** ± 0.011 |

Table S2: Per-class effect of the weighted sampler ($\gamma = 0.98$) relative to random sampling, on a selection of low-support or hard classes. $\Delta$ values are (weighted − random); positive favours the weighted sampler. "Support" is the test-set class size. The sampler ablation explored the grid $\gamma \in \{0.90, 0.95, 0.98, 0.99\}$ for the weight $w_c = K/(\gamma n_c + (1-\gamma)K)$, together with the alternative inverse-frequency parameterisation $w_c = (K/n_c)^{1/\alpha}$ for $\alpha \in \{1, 2, 3\}$; $\gamma = 0.98$ was retained for the final configuration.

| **Class** | **Support** | **ΔF1 (val)** | **ΔRec (val)** | **ΔF1 (test)** | **ΔRec (test)** |
|---|---|---|---|---|---|
| C21 | 6 | −0.0065 | +0.0476 | +0.0310 | 0.0000 |
| C1 | 10 | +0.1039 | +0.1143 | +0.0820 | +0.0857 |
| C23 | 10 | +0.0248 | +0.0714 | −0.0280 | +0.0429 |
| C9 | 27 | +0.0107 | +0.0106 | +0.0082 | 0.0000 |
| C2 | 30 | +0.0538 | +0.0952 | +0.0361 | +0.0809 |
| C17 | 37 | −0.0166 | +0.0152 | −0.0115 | +0.0425 |
| C22 | 37 | −0.0167 | +0.0116 | −0.0185 | 0.0000 |
| C7 | 38 | +0.0439 | +0.0445 | −0.0039 | +0.0110 |
| C13 | 42 | +0.0056 | +0.0102 | −0.0104 | −0.0170 |
| C5 | 53 | −0.0029 | −0.0054 | −0.0185 | −0.0243 |
| C6 | 55 | −0.0011 | −0.0052 | +0.0058 | 0.0000 |
| C24 | 56 | −0.0124 | +0.0332 | −0.0065 | +0.0434 |
| C18 | 58 | −0.0168 | 0.0000 | +0.0055 | +0.0351 |
| **Mean** | — | **+0.0130** | **+0.0341** | **+0.0055** | **+0.0231** |

Table S3: Effect of the number of selected input genes ($n$) on the proposed model (mean ± s.d., eleven seeds on the test set). $p$-values are from Wilcoxon signed-rank tests against the best configuration, GCN (700), with Holm correction for multiple comparisons. Best per column in bold. Performance peaks at 700 genes and plateaus between 500 and 1,000, indicating that the reported gain does not stem from simply enlarging the graph.

| **Model ($n$ genes)** | **Accuracy** | **F1 Macro** | **F1 Weighted** | $p$ | $p_{\mathbf{Holm}}$ |
|---|---|---|---|---|---|
| GAT (300) | $0.8060 \pm 0.0127$ | $0.7635 \pm 0.0152$ | $0.8009 \pm 0.0121$ | $4.9 \times 10^{-4}$ | 0.0034 |
| GCN (300) | $0.8101 \pm 0.0140$ | $0.7722 \pm 0.0222$ | $0.8087 \pm 0.0134$ | $3.4 \times 10^{-3}$ | 0.0171 |
| GAT (500) | $0.8229 \pm 0.0128$ | $0.7813 \pm 0.0180$ | $0.8198 \pm 0.0126$ | 0.0415 | 0.1245 |
| GCN (500) | $0.8283 \pm 0.0117$ | $0.7901 \pm 0.0169$ | $0.8252 \pm 0.0126$ | 0.3188 | 0.5195 |
| GAT (700) | $0.8200 \pm 0.0113$ | $0.7824 \pm 0.0155$ | $0.8188 \pm 0.0110$ | $6.8 \times 10^{-3}$ | 0.0273 |
| GCN (700) | $\mathbf{0.8285 \pm 0.0134}$ | $\mathbf{0.7918 \pm 0.0156}$ | $\mathbf{0.8264 \pm 0.0121}$ | — | — |
| GAT (1000) | $0.8038 \pm 0.0138$ | $0.7641 \pm 0.0162$ | $0.8017 \pm 0.0146$ | $9.8 \times 10^{-4}$ | 0.0059 |
| GCN (1000) | $0.8248 \pm 0.0140$ | $0.7864 \pm 0.0159$ | $0.8220 \pm 0.0147$ | 0.2598 | 0.5195 |

Table S4: Top five transcription factors per iCluster subtype, ranked by one-vs-rest ANOVA F-score on the training split. For each subtype we list the five TFs with the highest F-score, alongside the dominant cell-of-origin lineage of that cluster (after Hoadley et al.). The top of the ranking is enriched for lineage-defining master regulators, recovered without any external gene-set annotation. Gene symbols follow HGNC nomenclature. The ten subtypes with the cleanest lineage signal are shown; the full 27-subtype ranking is released with the project code.

| **Subtype** | **Lineage** | **Top TFs (one-vs-rest F-score, descending)** |
|---|---|---|
| C11 | Glioma (LGG) | OLIG2, POU3F2, NEUROD1, VDR, FOXG1 |
| C19 | Breast (luminal) | LMX1B, GATA3, TRPS1, TFAP2B, ESR1 |
| C4 | Pan-GI (colorectal) | CDX1, CDX2, NR1I2, PDX1, HNF4A |
| C26 | Liver (LIHC) | NR0B2, NR1H4, MLXIPL, NR5A2, NR1I2 |
| C28 | Pan-kidney | PAX2, NR1H4, HNF1B, HOXA4, HNF1A |
| C15 | Melanoma (SKCM/UVM) | PAX3, SOX10, FOXD3, ELF3, IRF6 |
| C6 | Ovarian (serous) | WT1, SOX17, PHOX2A, HOXD3, HOXD1 |
| C22 | Germ-cell (TGCT) | LIN28A, POU5F1, SALL4, GBX2, NR5A1 |
| C10 | Pan-SCC (squamous) | TP63, FOXE1, FOXN1, SOX2, OTX1 |
| C16 | Prostate (PRAD) | HOXB13, BHLHA15, GATA2, SIM2, SPDEF |

Table S5: Final MOGNN-TF configuration (Optuna trial 18, selected by validation macro-F1). The full search grid is reported in the project repository and reproduced verbatim by `config_final.yml`.

| **Hyperparameter** | **Value** |
|---|---|
| *Graph and feature selection* | |
| Selected genes ($N_g$) | 700 |
| Selected miRNAs ($N_m$) | 100 |
| Selected TFs ($N_t$) | 200 |
| Feature selection | variance |
| Active edge types | $E_{GG}, E_{MG}, E_{MM}, E_{TG}, E_{TT}, E_{TM}$ |
| *Architecture* | |
| GNN backbone | GCN |
| Pool size | 16 |
| Jumping knowledge | disabled |
| Parallel branch | enabled |
| Decoder (reconstruction) | enabled |
| Dropout | 0.2 |
| *Optimisation* | |
| Optimiser | AdamW |
| Learning rate | $8 \times 10^{-4}$ |
| Weight decay | $1 \times 10^{-4}$ |
| Manual L2 on GNN weights | $5 \times 10^{-4}$ |
| Batch size | 64 |
| LR schedule | cosine annealing |
| Max epochs / patience | 100 / 10 |
| *Class imbalance* | |
| Sampler | `WeightedRandomSampler` |
| Weighting | $w_c = K/(\gamma n_c + (1-\gamma)K)$ |
| $\gamma$ | 0.98 |
| Cross-entropy weighting | disabled (handled by sampler) |

Table S6: Effect of the training fraction on the proposed model (mean ± s.d., eleven seeds on the test set). The validation set is held at 10% and the remainder assigned to test. Performance degrades gracefully as supervision is reduced: GCN macro-F1 moves from 0.7978 at 80% to 0.7779 at 50%, and the GCN–GAT ordering is preserved at every fraction. The degradation is not statistically resolvable in the upper range; the 80% versus 60% GCN difference is not significant (Wilcoxon signed-rank, $p_{\text{Holm}} = 0.83$).

| **Train %** | **Model** | **Accuracy** | **F1 Macro** | **F1 Weighted** |
|---|---|---|---|---|
| 80% | GAT | $0.8287 \pm 0.0094$ | $0.7891 \pm 0.0205$ | $0.8255 \pm 0.0108$ |
| | GCN | $0.8355 \pm 0.0123$ | $\mathbf{0.7978 \pm 0.0205}$ | $0.8342 \pm 0.0119$ |
| 70% | GAT | $0.8181 \pm 0.0164$ | $0.7798 \pm 0.0216$ | $0.8153 \pm 0.0171$ |
| | GCN | $0.8245 \pm 0.0132$ | $0.7865 \pm 0.0155$ | $0.8224 \pm 0.0119$ |
| 60% | GAT | $0.8156 \pm 0.0141$ | $0.7756 \pm 0.0131$ | $0.8133 \pm 0.0139$ |
| | GCN | $0.8292 \pm 0.0144$ | $0.7971 \pm 0.0162$ | $0.8280 \pm 0.0142$ |
| 50% | GAT | $0.8106 \pm 0.0178$ | $0.7735 \pm 0.0187$ | $0.8091 \pm 0.0162$ |
| | GCN | $0.8169 \pm 0.0151$ | $0.7779 \pm 0.0178$ | $0.8154 \pm 0.0150$ |